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1. Introduction

1.1 Company Overview:



Motivate is a pioneer in digital manufacturing innovation, specializing in Vue.js applications tailored to streamline factory operations. Their core mission revolves around revolutionizing traditional factory setups into efficient, paperless environments while providing comprehensive insights for enhanced communication and a motivated workforce. Central to Motivate's offerings are the Motivate | OEE and Motivate | Shopfloor Control systems, which cater to line production and assembly/machine-based production, respectively. These systems are designed to optimize overall equipment effectiveness and enhance shop floor control through real-time data analysis and digitalization of critical processes.

1.2 Project Overview:

The project within Motivate's Platform Management Tool (PMT) is to develop an intuitive Instructions Creation Tool and enhance the quality of the PDF viewer. Currently, managers and operators need help with their reliance on external files. This project aims to integrate a solution directly into PMT, enabling users to create instructions on the platform, thus improving efficiency and user experience. My role involves refining the PMT interface to be more user-friendly and facilitating easy instruction creation, access, and management. Enhancing the HD PDF viewer within the iframe to significantly improve the user experience is also a key focus. Seamless integration of high-definition PDFs into the platform will allow users to access and utilize them directly within PMT. These enhancements are expected to streamline workflow and better meet clients' diverse needs, thereby increasing productivity in factory settings.

Main Research Question:

How can the user experience (UX) and user interface (UI) design of the PMT platform be optimized for efficient instruction management, while ensuring compliance with standardization requirements?

Sub-Research Questions:

1. What are the specific standardization requirements and industry standards applicable to instruction management within manufacturing environments?
2. How can the PMT's UI facilitate the intuitive creation and management of work instructions?
3. What workflow and sequence of steps are essential for effective instruction management in PMT?
4. What are the different types of work instructions encountered in manufacturing, and how can PMT accommodate them?
5. What do current competitive solutions offer for instruction management, and where do they fall short?
6. How do the environmental contexts in which instructions are used affect the design of PMT?
7. What platforms will interface with the PMT, and how can the system deliver uniform usability regardless of the platform?
8. What are the technical challenges and considerations in developing a PDF viewer feature that meets user needs and standardization requirements?

9. How can user feedback and iterative design processes be utilized to ensure the PDF viewer meets both usability and standardization criteria?

2. Assignment Overview

2.1 Context:

The PMT project addresses the challenge of reliance on external files for instructions, aiming to integrate a solution directly into PMT. Motivate specializes in creating digital solutions that revolutionize traditional manufacturing processes, making them more efficient and paperless. The project is significant as it aligns with Motivate's mission to enhance operational efficiency and provide actionable insights that drive productivity and motivation among factory workers.

2.2 Initial Situation:

Initially, the PMT needed an integrated tool for creating and managing instructions, leading to inefficiencies. Managers and operators had to rely on external files, which complicated the process and could have improved productivity. The absence of a unified platform for instruction management meant that users faced challenges in accessing, creating, and managing instructions efficiently.

2.3 Goals:

The primary goal is to develop a CRUD frontend for PMT, enabling efficient creation, viewing, and updating of instructions. Short-term goals include designing and implementing an intuitive interface that supports CRUD operations and enhancing the PDF viewer for better usability. Long-term goals involve integrating these solutions into the existing PMT platform, ensuring cross-browser compatibility, and providing comprehensive documentation for future maintenance.

3. Research and Analysis

3.1 Online Research:

Extensive online research was conducted to identify common UI/UX challenges associated with instruction creation tools in a factory context. The study focused on accommodating diverse instruction formats and addressing environmental factors that affect usability.

- Manufacturing work instructions have traditionally relied on various formats, including paper manuals, PDFs, images, slideshows, and videos. However, our standardized instruction builder within the PMT is designed to surpass the need for these disparate forms. Integrating all instructional content directly into the PMT ensures that users have seamless access to the instructions they need, regardless of the format. This consolidation makes it easier for users to navigate and comprehend the instructions, enhancing operational efficiency and ensuring clarity across all tasks. Our solution simplifies the user experience and supports diverse operational needs by providing a unified, intuitive platform for all instructional materials.
- Addressing Environmental Contexts in PMT Design: The design and functionality of PMT are significantly influenced by the environmental context in which they are deployed. Factors such as ambient noise, lighting conditions, and the physical layout of the workspace necessitate adaptable and thoughtful design solutions. For environments with visual distractions or noise, PMT must offer alternative means of presenting information, such as visually engaging PDFs or step-by-step image guides, to ensure uninterrupted access to critical information.
- Main Points from Online Research:
 - Digital formats for work instructions (PDFs, documents, images, slideshows) cater to diverse learning styles and operational needs.
 - Environmental factors like lighting and workspace layout necessitate adaptable UI/UX design to ensure the accessibility and usability of PMT.
 - Making instructional content easily navigable within PMT enhances user experience, allowing quick access to relevant information.
 - Adapting PMT design to include non-auditory instruction formats ensures uninterrupted access to critical information in noisy environments.
 - Integrating AI for spell-checking and writing assistance improves the quality and clarity of instructional content.

- Using innovative technologies, such as wearable trackers, introduces new possibilities for interactive and efficient task management [**Online research UI UX Challenges of factory context**].

3.2 Competitor Analysis

Competitor analysis helps understand the landscape of existing instruction creation tools and identify their strengths and weaknesses. This analysis provided valuable insights that informed the design and development of the new instruction management tool.

Main Takeaways:

- **Ease of Use:** Simplicity and ease of use, especially on mobile platforms, are crucial. Tools that offer a straightforward, intuitive interface are preferred.
- **Quick Edits and Feedback:** The ability to quickly create, edit, and receive feedback on instructions is highly valued.
- **Device Compatibility:** Compatibility with various devices, particularly mobile, is essential for usability in diverse environments.
- **Modern Design:** There is an opportunity to innovate with a more modern, personalized design, avoiding the impersonal or outdated feel of some existing tools.
- **Multilingual Support:** Supporting multiple languages is important for global usability.

Companies Analyzed:

During my research, I looked at several competitors:

- **SwipeGuide:** Offers multilingual support, quick creation and editing features, and a clean, simple design. However, it includes some gamified elements that may feel impersonal and a somewhat crowded interface.
- **Workclout:** Features a high-quality, intuitive design that is easy to read and follow, and is compatible with all devices. However, the call-to-action elements could be improved.
- **Poka:** Provides features for troubleshooting and user feedback tracking, but it needs more user engagement and feels complete compared to other tools.
- **Zaptic:** It has a good design for tablets and PCs, but lacks focus on mobile platforms and instructions are not the main focus.

These insights have been instrumental in shaping the design and functionality of the new instruction management tool, ensuring it meets user needs effectively and stands out in the competitive

landscape. The goal is to create a tool that is not only functional and efficient but also engaging and user-friendly [**Competitor Analysis Instruction Creation**].

3.3 Customer Interviews:

Interviewed key stakeholders, including managers and operators, to gather insights into their needs and challenges with the current system. These interviews revealed pain points such as difficulty in managing instructions, reliance on external files, and the need for a more integrated solution. The feedback was crucial in shaping the project requirements and guiding the design process.

Interview Highlights:

- **Critical Role of Instructions:** Instructions are essential for learning how to use machinery and perform tasks correctly. They are particularly valuable for new operators who are beginning to familiarize themselves with the equipment.
- **Value of Visual Aids:** Visual aids, including videos and photos, are highly valued by users. These aids help clarify complex instructions and provide a visual reference that enhances understanding.
- **Device Usage:** Tablets are the primary devices used for accessing instructions on the shop floor. However, users reported that these tablets are often slow, which can hinder efficiency.
- **Feedback Mechanisms:** There is a strong need for integrated feedback mechanisms and reporting features. Users expressed the importance of being able to provide feedback on instructions and report issues directly through the platform.
- **Need for Integration:** The interviews highlighted the need for a seamless and integrated solution that consolidates all instruction-related activities within the PMT. Users emphasized the importance of having a single platform to create, manage, and access instructions.
- **Reporting and Updating:** Users pointed out that incorporating reporting features could streamline problem-solving as issues arise. They also noted the importance of being able to update standards quickly to reflect changes in procedures or machinery.

The feedback from these interviews was instrumental in identifying the key requirements for the instruction management tool. It ensured that the tool would address the specific needs and challenges faced by users, ultimately leading to a more effective and user-friendly solution. These insights guided the initial design and laid the groundwork for subsequent development phases, ensuring that the final product would be well-aligned with user expectations and operational requirements [**Interview**].

3.4 Observational Studies:

Conducted observational studies to understand how factory workers interact with the current PMT system. These studies provided valuable insights into the practical challenges faced by users, such as navigating the interface, accessing instructions, and managing updates. Observations were used to identify areas for improvement and ensure that the new tool would meet real-world needs.

- **Technical Limitations:** Observational studies highlighted the need for responsive design to accommodate different devices and screen sizes. Using libraries like pdf.js or Vue-pdf was suggested to ensure customization and compatibility with various media formats. The system must handle different file types and resolutions effectively. Ensuring scalability as content increases is crucial for long-term success. The platform must be responsive across mobile devices and tablets to ensure accessibility in various factory settings.
- **Physical Observations:** Observations in the factory revealed that the working environment is often noisy, and operators rely on visual and auditory cues. Tablets are commonly used but are slow, highlighting the need for a responsive and efficient solution. Standardization of work practices, as promoted by Training Within Industry (TWI) methods, is essential for minimizing errors and enhancing productivity **[Observational Studies]**.

3.5 Conclusion:

The research and analysis phase underscored the necessity for an integrated instruction management tool within the PMT. Key findings highlighted the importance of user-friendly design, efficient functionality, and robust support for creating, viewing, and updating instructions. These insights informed the subsequent stages of the project, ensuring that the developed solutions would effectively address user needs and enhance overall productivity.

4. User Requirements and Expectations

4.1 Features List:

Compiled a comprehensive list of required features based on research and user feedback. Key features include CRUD functionality for instruction management, an enhanced PDF viewer, responsive design, and seamless integration with the existing PMT platform.

Feature List Highlights:

- **Creation of Standards:** Ability to create, edit, and manage standards with options to upload photos and videos. Features like subtitles for videos are considered.
- **Standards Approval:** Processes for approving standards, including notifications and reminders for approvers, rejection reasons, and feedback fields.
- **Standards Visualization:** Features for visualizing standards with zoom and pan options, dashboards for managing standards, and filters for easy search and navigation.
- **Feedback on Standards:** Implementing a rating system, comment sections, and revision history tracking.
- **Reporting Issues:** Functionality for uploading issues with photos/videos and a form for comments.
- **Standards Revision:** Options to edit standards without remaking them entirely, preserving legacy versions, and resubmitting standards for approval after changes.
- **Training and Competency:** Integration of standards with training modules, tracking completion, and registering users as trained in specific standards [**STANDARD WORK Feature list**].

4.2 MOSCOW Method Analysis:



Applied the MOSCOW method to prioritize features based on their importance:

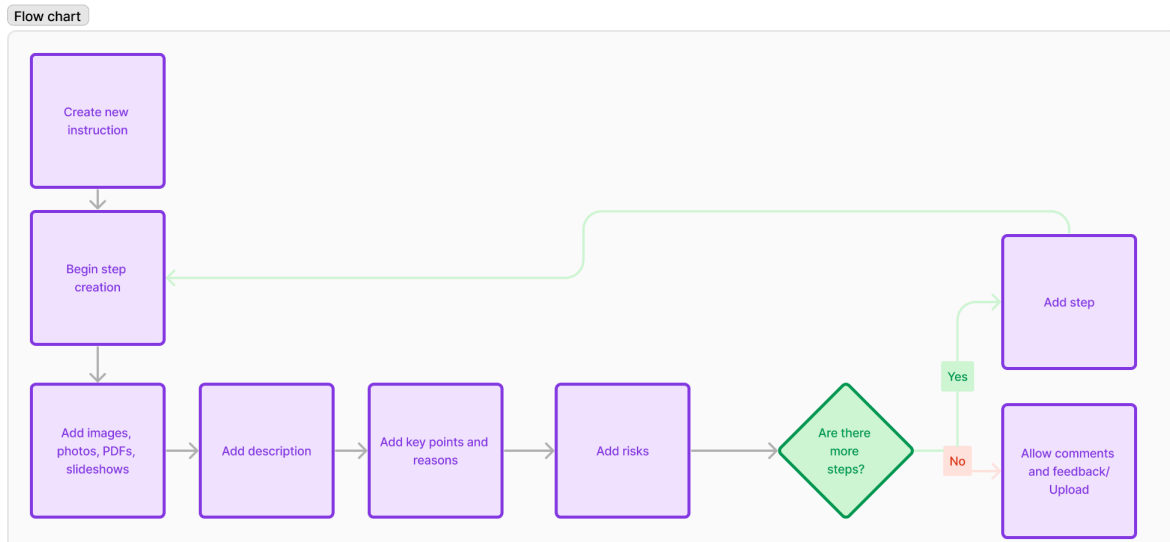
- Must-have: CRUD functionality, enhanced PDF viewer, responsive design, notifications for approval, and feedback mechanisms.
- Should-have: Advanced editing tools, real-time collaboration, rating system, search and filter functionality, and multilingual support.
- Could-have: AI-assisted text, PDF integration into the existing program, virtual reality training modules, complex analytics, and template libraries for standards.
- Won't-have: Features that do not add significant value at the moment or are out of scope, such as certain backend developments or low-priority functionalities **[MoSCoW]**.

4.3 Conclusion:

The features list and MOSCOW analysis provided a clear framework for developing the instruction management tool. Prioritizing features ensured that the most critical aspects were addressed first, while also outlining potential areas for future enhancement. This structured approach helped streamline the development process and ensure that the final product would meet user needs and expectations.

5. Prototyping and User Testing

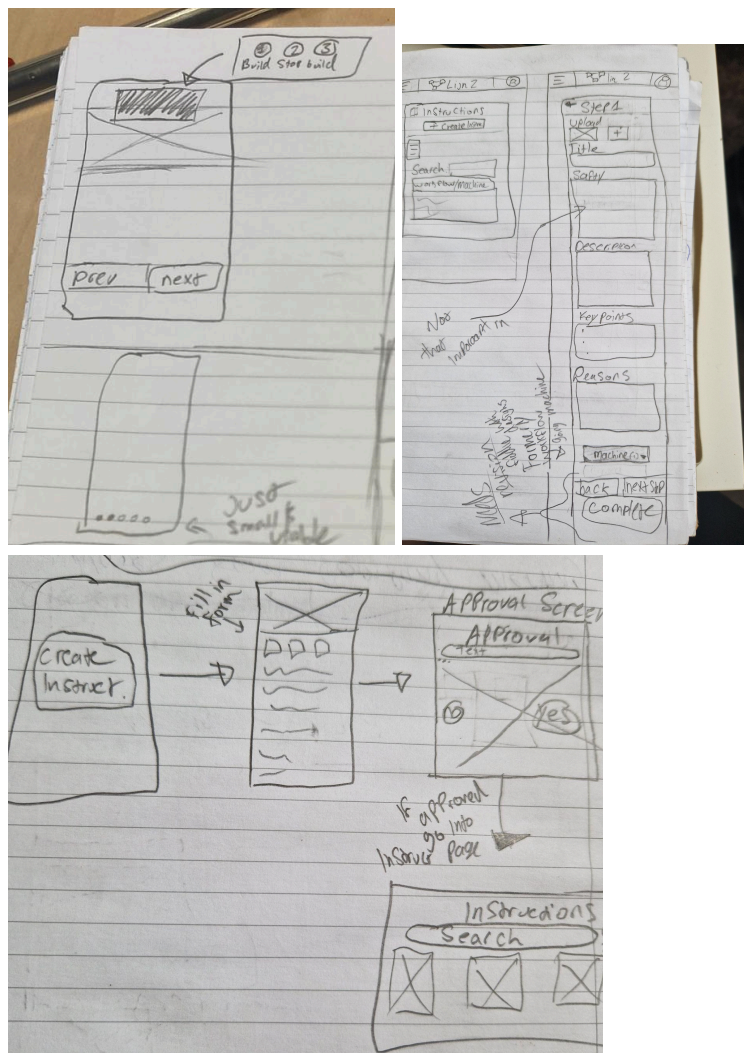
Workflow for Instruction Creation: To visualize the process, a workflow helps to outline the steps involved in creating new instructions following the TWI methodology. This workflow was a crucial part of the initial design phase, helping to map out the necessary steps and ensure all aspects of instruction creation were considered.



1. **Create new instruction:** Initiate the creation process.
2. **Begin step creation:** Start detailing the steps.
3. **Add images, icons, and videos:** Incorporate various media to support the instructions.
4. **Add description:** Provide a detailed description for each step.
5. **Add key points and reasons:** Highlight important points and reasons for each step.
6. **Are there more steps?:** Determine if additional steps are needed.
 - If yes, add another step.
 - If no, proceed to the final step.
7. **Allow comments and feedback/Upload:** Enable comments and feedback, then upload the instructions.

This workflow provided a clear guide for both the design and development phases, ensuring a comprehensive and user-friendly tool for instruction creation.

5.1 Low-Fidelity Prototypes:



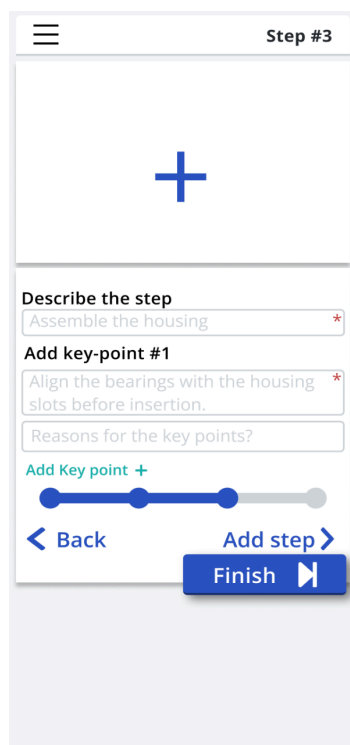
Initial prototypes based on research findings. These low-fidelity prototypes focused on the basic structure and functionality of the tool, providing a visual representation of the proposed design.

5.2 User Feedback on Low-Fidelity Prototypes:

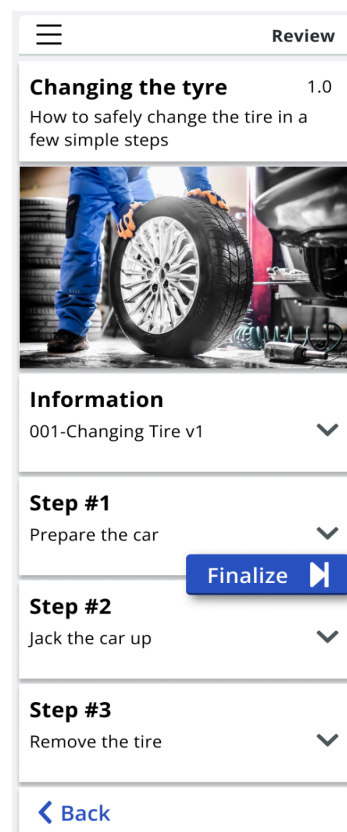
To gather feedback on the low-fidelity prototypes, initial user interviews were conducted. Users were shown the sketches and asked questions to ensure that all necessary information was covered and to assess the usability, design, and functionality of the proposed tool. This feedback was crucial in validating the direction of the design and confirming that the approach met user needs.

The feedback provided insights into the clarity of navigation, the intuitiveness of the design, and the overall user experience. Users validated that the proposed features and design elements were on the right track, reinforcing the need for a user-friendly interface and robust functionality. This validation was essential in ensuring that the design aligned with user expectations and laid a solid foundation for further development and refinement.

5.3 High-Fidelity Prototypes:



The screenshot shows a mobile app interface for 'Step #3'. At the top is a hamburger menu icon and the title 'Step #3'. Below is a large blue plus sign in a white box. Underneath is a section titled 'Describe the step' with a text input field containing 'Assemble the housing' and a red asterisk. Below that is 'Add key-point #1' with a text input field containing 'Align the bearings with the housing slots before insertion.' and a red asterisk. Below that is a text input field for 'Reasons for the key points?'. At the bottom left is a blue 'Back' button with a left arrow. At the bottom right is a blue 'Add step' button with a right arrow. In the center bottom is a blue 'Finish' button with a play icon. Above the 'Finish' button is a progress indicator with a blue line and two dots, and a green 'Add Key point +' link.



The screenshot shows a mobile app interface for 'Review'. At the top is a hamburger menu icon and the title 'Review'. Below is a section titled 'Changing the tyre' with a version number '1.0' and a subtitle 'How to safely change the tyre in a few simple steps'. Below that is a photo of a person changing a tire. Underneath is an 'Information' section with the text '001-Changing Tire v1' and a dropdown arrow. Below that is a 'Step #1' section with the text 'Prepare the car' and a dropdown arrow. Below that is a 'Step #2' section with the text 'Jack the car up' and a dropdown arrow. Below that is a 'Step #3' section with the text 'Remove the tire' and a dropdown arrow. At the bottom left is a blue 'Back' button with a left arrow. At the bottom right is a blue 'Finalize' button with a play icon.

Designed high-fidelity prototypes based on feedback from low-fidelity prototypes. These detailed, interactive prototypes incorporated refined design elements and enhanced functionality, providing a more accurate representation of the final product.

5.4 User Testing on High-Fidelity Prototypes:



Conducted comprehensive user testing on high-fidelity prototypes. Users provided detailed feedback on the usability, effectiveness, and overall experience of the tool. This iterative testing process helped ensure that the final design met user needs and expectations.

Main Takeaways from User Tests:

- **Ease of Use and Design:**
 - Users consistently praised the design and overall concept, finding it intuitive and straightforward.
 - The simplicity and clarity of the interface were highlighted as strong points.
- **Functionality Enhancements:**
 - There was a clear demand for responsive design, enabling instruction creation and management on both PCs and mobile devices.

- Users suggested the addition of safety icons and input methods to enhance compliance with regulations.
- **Additional Features:**
 - The ability to export standards to PDF was a common request, accommodating regulatory requirements.
 - A review system and the inclusion of roles and notifications for standard approvals were seen as valuable additions.
- **Content Management:**
 - Auto-generated document numbers with editable options were suggested to streamline document management.
 - Multiple approval levels for standards were recommended to ensure thorough validation and compliance.
- **User Interaction:**
 - Periodic reminders to review and update standards were proposed to maintain relevance and alignment with company goals.

These insights were instrumental in refining the high-fidelity prototypes, ensuring that the final product would meet diverse user needs and expectations effectively.

5.5 Conclusion:

The prototyping and user testing phases were critical in refining the design and functionality of the instruction management tool. Iterative testing and feedback helped identify and address issues early, ensuring that the final product was user-friendly and effective. The feedback from high-fidelity prototype testing provided valuable insights into user preferences and areas for enhancement, leading to a more refined and comprehensive solution **[User Tests Sivomatic]**.

6. Implementation

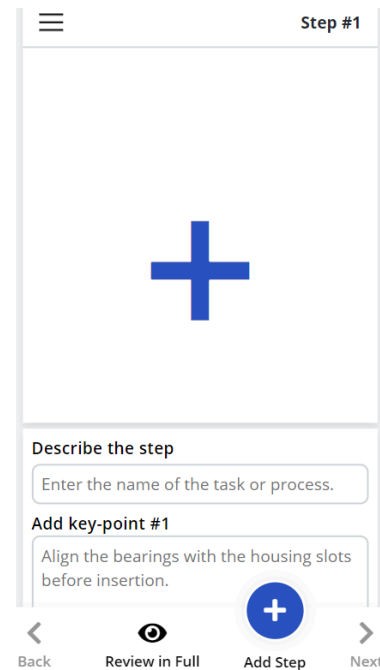
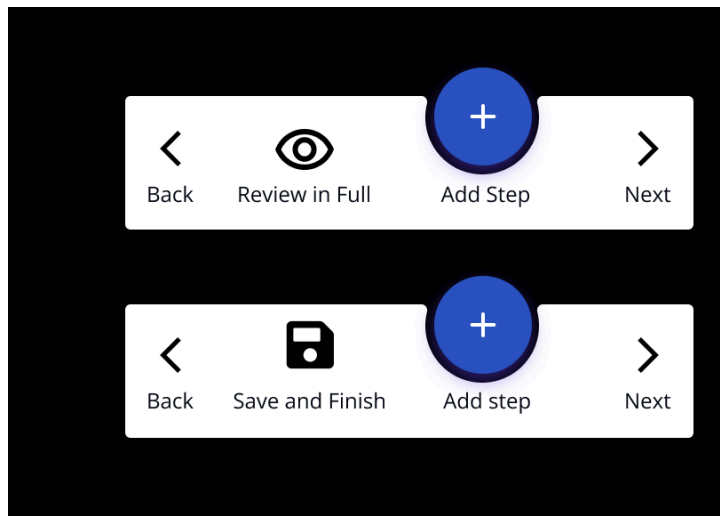
6.1 Development Process:

The development process for the instruction creation tool was challenging due to the necessity of integrating with the existing database to upload new types of instruction data while maintaining legacy systems. This included the continued use of PDF and image-viewing functionalities to preserve the company's existing standards in an updated viewer. Early in the project, the PDF viewer was enhanced to better understand the existing codebase and ensure seamless integration.

Subsequently, a comprehensive new standard builder was developed, allowing users to upload various types of data from mobile devices. This data can later be viewed and managed on PCs or tablets. The new system was designed using Vue.js, focusing on creating a responsive and user-friendly interface. The development involved iterative coding, extensive testing, and continuous refinement to meet the specified requirements.

However, the main issues encountered during implementation were related to connectivity, particularly with the uploading of images over slow data connections. This issue required rigorous testing and remains a work in progress to ensure reliable performance across different network conditions. The factory floor environment significantly impacts the application's usability, as factors such as intermittent connectivity, noise, and physical conditions can hinder efficient use. Ensuring the application functions effectively under these challenging conditions is paramount for its success.

6.2 User Testing & Changes:



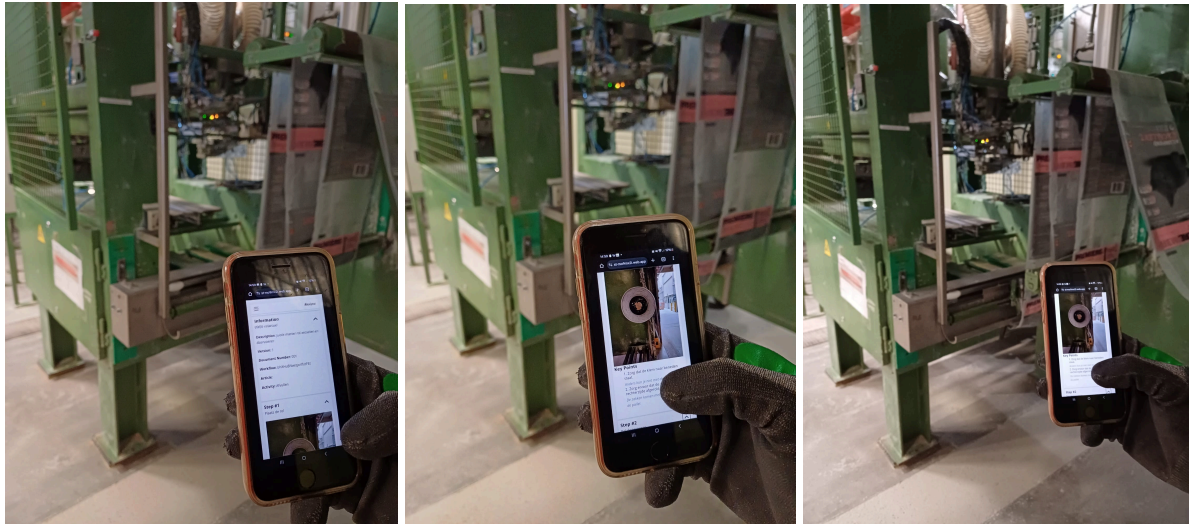
User testing involved operators creating instructions, requiring them to adopt TWI (Training Within Industry) methods. Testing revealed some design confusion, leading to the addition of a bottom navigation bar. Training for operators was identified as necessary.

Key takeaways included:

- **Bigger, More Obvious Buttons:** For intuitive navigation.
- **Red Arrows and Photo Editing Tools:** To enhance clarity.
- **Training and Examples of Good Standards:** Built into the application.
- **Safety Stickers and Warning Icons:** For quick risk communication.

These changes need to be made to improve user experience and effectiveness.

6.3 Front-End Performance Tests:



Front-end performance tests were conducted to evaluate the tool's responsiveness, speed, and overall performance across various browsers and devices. These tests aimed to ensure that the tool was performant and met all technical requirements. Key aspects tested included:

- **Responsiveness:** Ensuring the interface adapts seamlessly to different screen sizes and orientations.
- **Speed:** Optimizing loading times and minimizing latency, particularly for data-intensive operations like image uploads.
- **Cross-Browser Compatibility:** Verifying functionality and appearance across different web browsers to ensure a consistent user experience.
- **Environmental Impact:** Assessing the tool's performance in a factory setting, where connectivity can be unreliable and the physical environment can be challenging.

6.4 Conclusion:

The implementation phase successfully developed a fully functional CRUD frontend for the PMT. Performance tests confirmed that the tool was responsive, efficient, and met all specified requirements. The integrated instruction management tool significantly enhanced the usability and functionality of the PMT.

Despite these successes, the development process presented numerous challenges, particularly in ensuring seamless integration with legacy systems and managing data uploads in a complex factory environment. These challenges have provided invaluable learning experiences, allowing significant growth in understanding what it takes to develop a robust, real-life application.

While the tool works well overall, some issues remain, particularly with video uploads over slow connections. Continued efforts are needed to address these issues. However, other functionalities are performing well, and users on the shop floor have been able to effectively utilize the application, enhancing their productivity and efficiency.

7. Conclusion

7.1 Summary:

The project successfully developed an integrated instruction creation tool that significantly enhanced user experience and productivity within the PMT. The tool was designed to address the initial challenges of reliance on external files by providing a seamless, in-platform solution for creating, managing, and viewing instructions.

To ensure the tool met user needs, extensive research and analysis were conducted. This involved online research, customer interviews, and observational studies, providing a comprehensive understanding of the challenges users faced. The findings underscored the importance of a user-friendly design, efficient functionality, and robust support for various instruction formats. This foundational knowledge guided the development process and ensured the tool was built with the end-user in mind.

Next, a detailed list of required features was compiled based on the research insights. Using the MOSCOW method, these features were prioritized into must-haves, should-haves, could-haves, and won't-haves. This structured approach helped focus development efforts on the most critical aspects. All must-have features were successfully implemented, including CRUD functionality, an enhanced PDF

viewer, responsive design, and approval notifications with feedback mechanisms. This careful prioritization ensured that the tool addressed the core needs of its users effectively.

The prototyping and user testing phases were pivotal in refining the tool's design and functionality. Both low-fidelity and high-fidelity prototypes were developed, which were rigorously tested with users. The feedback gathered from these tests was invaluable. It led to significant improvements, such as the addition of a bottom navigation bar, more prominent buttons, red arrows, photo editing tools, and built-in training examples. These enhancements made the tool more intuitive and user-friendly.

The development process itself was challenging, especially integrating the tool with the existing database while maintaining legacy systems. Using Vue.js, the focus was on creating a responsive and user-friendly interface. One of the major hurdles was ensuring reliable data uploads, particularly in the complex factory environment where connectivity issues are common. Despite these challenges, the tool was developed through iterative coding, extensive testing, and continuous refinement.

The implementation phase confirmed that the tool was responsive, efficient, and met all specified requirements through comprehensive performance tests. Although some issues with video uploads over slow connections remain, other functionalities performed well. Users on the shop floor have effectively utilized the application, enhancing their productivity and efficiency.

Overall, the application has proven to be a valuable asset in real-life situations. The project has provided invaluable learning experiences, helping to understand what it takes to develop a robust, real-life application. While there is still some work to be done on improving video uploads, photo editing, and incorporating job safety stickers, these can be addressed in future updates. The application's success validates its potential to make workers' jobs easier and more efficient, ultimately contributing to increased productivity and a better working environment.

7.2 Challenges:

Throughout the project, several challenges were encountered, providing significant learning opportunities:

1. **Integration with Legacy Systems:** Integrating the new instruction creation tool with existing legacy systems was particularly challenging. Ensuring seamless compatibility while maintaining the functionality of the older systems required a deep understanding of both the new and existing codebases.
2. **Data Upload and Connectivity Issues:** The factory environment presented significant challenges, especially regarding data upload and connectivity. Slow connections impacted the

ability to upload videos and images efficiently. Addressing these connectivity issues required rigorous testing and optimization, which is an ongoing effort.

3. **Design Requirements:** Adhering to the company's specific programming and design requirements, as well as established design patterns, posed a significant challenge. Balancing these constraints while trying to innovate and improve the tool required meticulous planning and execution.
4. **Working with a Complex Platform:** Developing within a complex platform like PMT was demanding. Understanding the intricacies of the platform, ensuring all new components are integrated smoothly, and maintaining overall system stability required a high level of technical proficiency and problem-solving skills.
5. **Environmental Constraints:** The physical conditions of the factory, such as noise and lighting, impacted the usability of the tool. Designing a user interface that could be easily navigated in such conditions required thoughtful consideration and adaptability.

Despite these challenges, the successful implementation and positive user feedback have validated the tool's effectiveness and potential for improving operational efficiency on the shop floor. The continuous improvement process will address remaining issues, ensuring the tool evolves to meet user needs effectively.

7.3 Advice to Stakeholders

Continuous Improvement: Establish a continuous improvement process to ensure the Instruction Management Tool remains effective and relevant. Regularly collecting user feedback will provide valuable insights into how the tool is used in practice and highlight areas for enhancement. This feedback can be gathered through surveys, interviews, and usage analytics. By implementing a structured process for analyzing this feedback and updating the tool accordingly, the company can address new requirements and improve functionality over time. Regular updates will help maintain user satisfaction and ensure the tool evolves to meet changing needs.

Training and Support: To maximize the benefits of the Instruction Management Tool, provide comprehensive training and support for users. Training sessions should be conducted to familiarize users with the tool's features and best practices for creating and managing instructions. Additionally, support materials such as user manuals, video tutorials, and FAQs should be readily available. Offering ongoing support through helpdesks or online forums can help address any issues users encounter, ensuring they can fully utilize the tool's capabilities. Effective training and support will enhance user confidence and adoption, leading to better overall outcomes.

Enhanced Image Editing Capabilities: Users have highlighted the necessity for advanced image editing features within the tool. Integrating functionalities such as cropping, arrow addition, and other editing tools will enable users to distinctly emphasize critical aspects of images. These enhancements are pivotal in elevating the clarity and impact of the instructions.

Video Functionality: Improving video upload functionality is another critical area. Currently, issues with video uploads over slow connections can hinder the user experience. Addressing these issues by optimizing the video upload process and ensuring smooth playback will enhance the tool's usability. Consider implementing video compression techniques and adaptive streaming to improve performance across varying network conditions.

Explore Advanced Features: To further enhance the Instruction Management Tool's capabilities, consider adding advanced features such as analytics and multimedia support. Integrating analytics can provide valuable insights into how the tool is used, helping identify patterns and areas for improvement. For instance, tracking which instructions are accessed most frequently or where users encounter difficulties can guide future updates. Adding multimedia support, such as the ability to embed videos or interactive diagrams, can enrich the instruction content and improve user engagement. These advanced features can make the tool more powerful and versatile, better serving the needs of its users.

Possible AI Features: The integration of AI technologies into the Instruction Management Tool offers significant opportunities to enhance functionality and user experience. Potential AI applications include:

- **AI-Powered Feedback Analysis:** Implement natural language processing (NLP) algorithms to categorize and analyze user feedback, identifying common issues and suggesting improvements.
- **Intelligent Content Suggestions:** Utilize machine learning algorithms to analyze existing instruction content and user interactions, suggesting additional content or improvements to ensure comprehensiveness and clarity.
- **Automated Spell-Checking and Grammar Correction:** Integrate AI-based tools to ensure all instructional content is clear and error-free, improving professionalism and readability.
- **Predictive Maintenance Alerts:** Use AI to analyze usage patterns and predict when equipment might require maintenance, providing timely alerts to prevent unexpected downtime.
- **Enhanced Search Functionality:** Implement AI-driven search algorithms to provide more accurate search results based on context and relevance, helping users find relevant instructions quickly.

- **Personalized Learning Paths:** Leverage AI to create personalized learning paths for users based on their role, experience level, and interaction with the tool, enhancing learning outcomes and engagement.

By exploring these advanced features and potential AI applications, the company can significantly enhance the Instruction Management Tool's functionality, making it more efficient and user-friendly. As the tool continues to evolve, these enhancements will ensure it remains aligned with the needs of users and the operational goals of the company.

8. Reflection

This semester has been a transformative period of growth and learning, navigating various challenges, acquiring new skills, and contributing meaningfully to the development of the Instruction Creation Tool. Engaging deeply in the project's design, development, and user experience provided hands-on experience crucial for real-world applications. Balancing technical requirements with user needs ensured that the solutions developed were not only functional but also user-friendly and efficient. Personally, this experience has reinforced my passion for UI/UX development and highlighted the importance of adaptability and continuous learning.

The project required integrating previously acquired knowledge and skills in an authentic context, delivering relevant and impactful results. Extensive research, including online research, customer interviews, and observational studies, was conducted to understand the needs and pain points of the users. This process was instrumental in shaping the project requirements and guiding the design process. Personally, this research phase deepened my understanding of user-centered design and the importance of empathy in creating effective user interfaces.

The iterative development and user testing phases were critical in refining the tool's design and functionality. The importance of feedback and how to incorporate it constructively to improve the final product was highlighted. Leading the development of prototypes and user testing phases enhanced leadership skills and underscored the value of collaboration and targeted interaction with stakeholders.

This experience taught me the value of leadership and teamwork, essential skills for any professional setting.

Programming the application in a way that is easy to follow and aligns with the rest of the application ensures seamless integration and maintainability, which is crucial for the ongoing success and usability of the tool. This technical challenge sharpened my coding skills and highlighted the importance of writing clean, maintainable code. It also emphasized the need for foresight and planning in software development.

Learning Outcomes:

Professional Duties:

- Developed a fully functional CRUD frontend, including design and implementation. Led the development of prototypes and user testing phases, iteratively improving the design. Built a complete and working version of the instructions creation tool.
- **Reflection:** This project required applying technical skills in Vue.js and UI/UX design, ensuring the production of professional, high-quality work that met industry standards. By delivering a product that aligns with the professional requirements of the IT area, the ability to carry out professional duties effectively was demonstrated. These experiences have laid a solid foundation for becoming a professional UI/UX developer with a deep understanding of front-end development. Personally, this reaffirmed my commitment to pursuing a career in UI/UX development. The rigorous demands of the project also taught me the importance of diligence, precision, and a commitment to continuous improvement in my professional practice.

Situation-Oriented:

- Conducted comprehensive online research and customer interviews to understand user needs. Used observational studies to identify and solve usability issues in the PMT and conducted customer tests.
- **Reflection:** Understanding the user context and needs was critical to delivering relevant and practical solutions that enhanced the PMT. This experience improved my ability to adapt skills to real-world situations, ensuring that the solutions provided were both practical and effective. This skill is essential for a UI/UX developer, as it involves empathizing with users and designing interfaces that meet their needs. Personally, it was rewarding to see the direct impact of my work on improving user experience. This phase of the project also honed my research skills, enabling me to gather and analyze data more effectively to drive design decisions.

Future-Oriented Organisation:

- Integrated features that support future scalability and updates, ensuring long-term usability and efficiency. Programmed the application to be buildable and easily scalable for future developers, allowing for the implementation of recommended features and updates. Engaged with stakeholders through interviews and feedback sessions to ensure the project met user expectations.
- **Reflection:** The approach considered long-term impacts and sustainability, aligning with the company's future-oriented goals. The application was designed and programmed to be buildable and scalable, ensuring that future developers can easily add new features and improvements. This outcome taught me the importance of planning for the future and incorporating scalability into design and development processes. This strategic thinking is vital for career aspirations in UI/UX development, where long-term usability and efficiency are paramount. Personally, this has instilled a greater appreciation for sustainable development practices. Understanding how to future-proof a product has been a significant takeaway, emphasizing the importance of building solutions that are not just functional today but remain relevant and effective in the long term.

Investigative Problem Solving:

- Used observational studies to identify and solve usability issues in the PMT and conducted customer tests. Conducted comprehensive online research and customer interviews to understand user needs.
- **Reflection:** This project enhanced problem-solving skills, allowing critical analysis and addressing user pain points effectively. It emphasized the value of thorough research and iterative testing in developing practical solutions, ensuring that the final product was both user-friendly and efficient. Problem-solving is a core competency for a UI/UX developer, enabling the creation of intuitive and efficient user interfaces. Personally, it was gratifying to see the iterative improvements lead to a polished final product. This experience reinforced the importance of a methodical approach to problem-solving, where each challenge is an opportunity to learn and improve the design.

Personal Leadership:

- Led the development of prototypes and user testing phases, iteratively improving the design. Developed a fully functional CRUD frontend, including design and implementation. Took control of the project, actively working with the company to align with their goals and independently conducting research and design as needed.
- **Reflection:** Demonstrated leadership by taking ownership of the development process and ensuring continuous improvement through user feedback. Engaging in entrepreneurial

activities around projects and focusing on personal development helped me grow as an IT professional and prepared me for future leadership roles. This experience is crucial for a UI/UX developer, as it involves leading design projects and collaborating effectively with cross-functional teams. Personally, it has boosted my confidence in leading and managing projects. Additionally, guiding a team through the iterative process of prototyping and testing emphasized the importance of fostering a collaborative environment, where everyone's input is valued and contributes to the project's success. My proactive approach in taking control of the project and independently driving research and design efforts further solidified my leadership capabilities and independence.

Targeted Interaction:

- Engaged with stakeholders through interviews and feedback sessions to ensure the project met user expectations. Conducted comprehensive online research and customer interviews to understand user needs.
- **Reflection:** Effective communication and collaboration with stakeholders were essential for the project's success, highlighting the importance of targeted interaction. By identifying key partners and working constructively with them, the project achieved its desired impact. I needed to talk to everyone who would use the product, from shop floor workers to managers, to see all points of view and fully actuate my project. Additionally, I listened to feedback from the internship company, which had its own wants and needs. Strong communication skills are essential for a UI/UX developer, facilitating collaboration and ensuring that design solutions meet user and business needs. Personally, this experience improved my ability to articulate ideas, gather valuable feedback, and balance diverse perspectives. This broadened my perspective on the importance of understanding the varied needs and expectations of different user groups to create a well-rounded product.

Overall, this internship has been instrumental in my professional development. It provided practical experience in applying ICT skills, reinforced the importance of stakeholder engagement, and emphasized the value of continuous improvement and adaptation. These experiences have equipped me with the knowledge and skills necessary for a successful career as a UI/UX developer with a deep understanding of front-end development.
